

Lab 3: SQL for Finance Data

1. Lab Overview

This lab uses SQL to query, audit and clean financial records in the NovaCore database. Use this README to check the ZIP contents, set up your working environment and troubleshoot common issues before starting the lab notebook.

2. Included in the ZIP Package

Your lab package contains everything required to complete the session. Please ensure you have the following files extracted before beginning:

A. Data folder

The **Data** folder contains the database and supporting database diagram:

- **novacore_lab3_database.duckdb:** The primary DuckDB database file containing all required tables and records for your analysis.
- **Database ERD:** A visual map of the NovaCore database structure, illustrating the links between customers, invoices, payments, and products.

B. Notebooks folder

The **Notebooks** folder contains the notebook files used for the lab:

- **SQL_Lab3.ipynb:** Primary lab notebook. This is your **main** workspace for the session.
- **Database_Setup.ipynb:** Optional setup notebook. Use this only if instructed or if you experience issues loading the provided database file. This notebook can create the database for you. You can then download the database and upload it to the SQL_Lab3 notebook.

C. Results folder

The **Results** folder is provided for any outputs, screenshots, exported files, or completed work generated during the lab.

D. Instruction PDF

- **Lab 3 Instructions.pdf:** Your comprehensive guide covering learning outcomes, study materials, and industry-focused tips.

E. Read-only walkthrough

- **Lab 3 Readonly.pdf:** A read-only walkthrough showing the practical process and expected outputs.

Note: Depending on your device settings, file extensions such as **.ipynb**, **.duckdb**, **.pdf**, or **.jpg** may or may not be visible. Use the file names and folders above to identify the correct files.

3. Important notice: third-party software and tools

These labs are designed to be completed on your own device using your choice of suitable software. You should use a desktop or laptop computer to download and undertake the labs. Mobile phones and other mobile devices should not be used for downloading or undertaking the labs.

ACCA provides the lab instructions and anonymised datasets only. ACCA does not provide, manage, license, maintain, endorse, or provide technical support for any third-party software, websites, platforms, plug-ins, extensions, or services referenced in these materials.

Any tools listed in this lab package are provided as examples of commonly used tools that may be suitable for completing the activities. You may use these tools or an equivalent alternative, subject to availability in your location, your device specification, your internet access, and any policies set by your employer, university, or local jurisdiction.

If you choose to use a third-party tool, you are responsible for:

- ensuring that the tool is appropriate for your device and operating system;
- reviewing and accepting the vendor's licence terms, privacy notice, security settings, and usage conditions;
- obtaining any required licence or account;
- downloading or accessing the tool only from the vendor's official website;
- resolving installation, access, account, compatibility, or technical issues directly with the relevant vendor or provider.

ACCA is not responsible for any loss, damage, data exposure, malware, service interruption, licence cost, or technical issue arising from the use of third-party tools or unofficial download sources.

ACCA Customer Services can support issues relating to ACCA-provided lab files, such as access to the ZIP file, missing files, corrupted downloads, or unclear ACCA instructions. ACCA Customer Services cannot provide support for installing, configuring, troubleshooting, licensing, or using third-party software.

4. Example compatible tools and official vendor links

To ensure you have the flexibility to work in the environment that best suits your existing workflow or institutional requirements, you may utilise the following platforms. While the lab is designed for portability, please note the specific configurations for each.

Compatible Platforms & Tools

Category	Tool/Platform	Notes for Students
Cloud notebook	Google Colab	Recommended example environment. Runs in a browser and normally requires no local Python installation.
Browser-based notebook	Try JupyterLab	Alternative tested with this lab. Upload the notebook and database in the required folder structure.
Local notebook	Jupyter Notebook	Suitable if you already use Python on your computer. You may need to install the required packages.
Cloud platform	Baidu AI Studio	Possible cloud alternative. Check access and package compatibility before use.
Development environment	Visual Studio Code	If you already use VS Code, you can open the Lab 3 notebook with the Jupyter extension and select a Python environment with the required packages.
Required database	DuckDB	Required for Lab 3. It runs through Python, you do not need a separate database server.

Optional local setup: [Anaconda Distribution](#) provides Python and Jupyter tools if you prefer to work on your computer.

5. Safe software access guidance

To reduce the risk of downloading unsafe or unauthorised software:

- use only official vendor websites;
- avoid download links from adverts, forums, file-sharing sites, unofficial mirrors or unknown sources;
- check the website address carefully before downloading;
- do not disable antivirus, firewall, browser or device security controls to install software;
- do not install software on employer, client or university devices unless permitted by local policy;
- do not upload personal, confidential, client, employer or assessment data into third-party tools;
- use only the anonymised datasets provided in the ACCA lab package.

6. What ACCA supports and does not support

Issue	ACCA support?	Route
Cannot access StudyHub or lab page	Yes	ACCA Customer Services
Cannot download the ZIP file	Yes	ACCA Customer Services
ZIP file appears corrupted or incomplete	Yes	ACCA Customer Services
ACCA-provided PDF, CSV or notebook file is missing	Yes	ACCA Customer Services
Lab instruction appears unclear or inconsistent	Yes, through defined learning/content support route	ACCA support route
Installing, configuring or troubleshooting Excel, Power BI, Tableau, Python, Jupyter or other third-party tools. Where DuckDB is used, it is embedded in the provided Python notebook and no separate DuckDB installation is expected.	No	Vendor or local IT support
Vendor account, login or password issue	No	Vendor support
Licence, subscription or trial issue	No	Vendor support
Employer/university device restrictions	No	Employer/university IT
Antivirus, firewall or operating system issue	No	Local IT or vendor support
Student cannot or does not wish to use tools	Not a technical support issue	Use Virtual Walkthrough

7. How to Start the Lab

Before you begin, read **Lab 3 Instructions.pdf**. This document explains the learning objectives, the tasks you will complete, and how the lab aligns with the ACCA syllabus.

This lab can be completed using any environment that supports Jupyter notebooks, Python, and DuckDB. Throughout this guide, Google Colab is used as an example because it is free, runs in a web browser, and requires no software installation. If you prefer, you may use another compatible environment, such as Jupyter Notebook, JupyterLab, VS Code, or Anaconda.

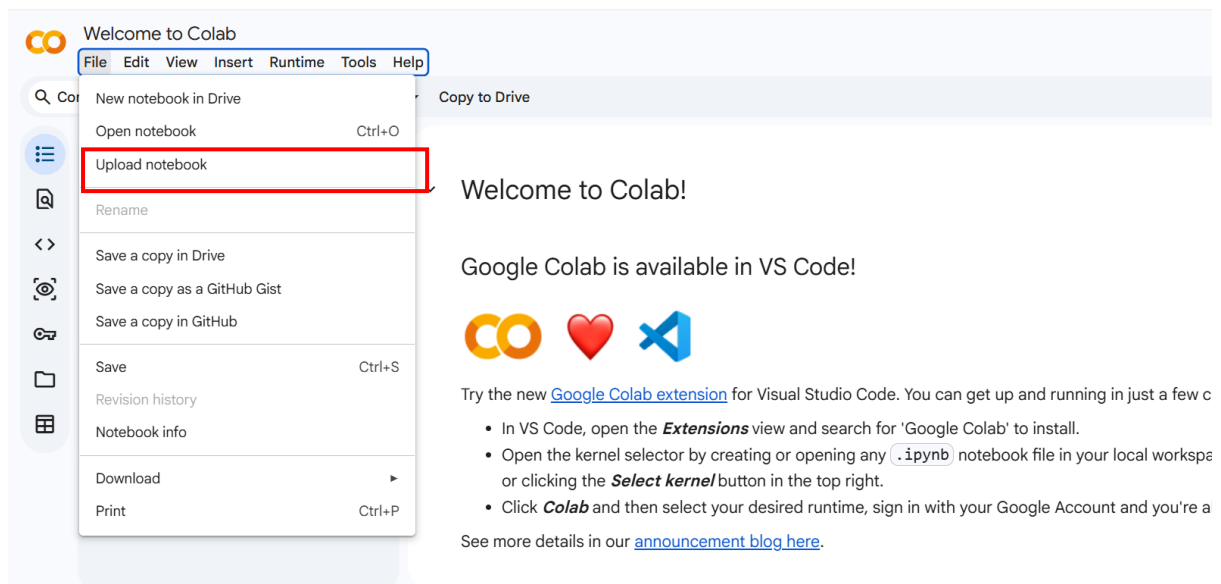
The overall workflow is the same: **open the notebook, make the database available to your environment, and run the notebook cells in order.**

Example: Using Google Colab

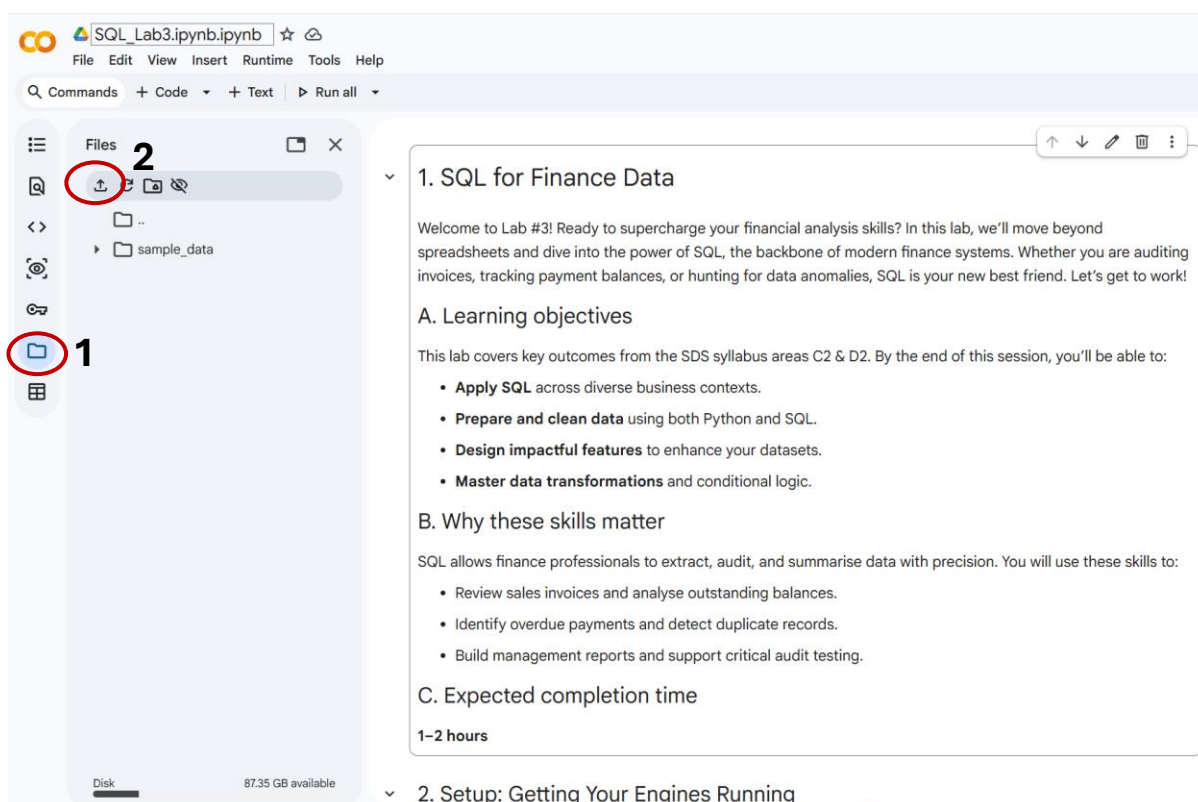
The following steps show how to complete the lab using Google Colab. If you are using a different environment, the overall process is similar.

Before starting, make sure you are signed in with a Google account. If you do not have a Google account, you will need to create one to use Google Colab and Google Drive.

1. Go to <https://colab.research.google.com> and upload the SQL_Lab3 notebook from the lab folder.



2. Wait a few seconds for the notebook environment to load. Then upload the novacore_lab3_database database file from the lab folder to the Colab file pane.



The screenshot shows the JupyterLab interface. On the left, the 'Files' sidebar is open, displaying a file explorer. A red circle labeled '1' highlights the 'upload' icon (a square with an upward arrow) in the sidebar. Another red circle labeled '2' highlights the 'upload' icon in the top bar of the file explorer. The main notebook area shows the title 'SQL_Lab3.ipynb.ipynb' and a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu bar is a search bar and a 'Run all' button. The notebook content is titled '1. SQL for Finance Data' and includes a welcome message, learning objectives, and a section on why these skills matter.

1. SQL for Finance Data

Welcome to Lab #3! Ready to supercharge your financial analysis skills? In this lab, we'll move beyond spreadsheets and dive into the power of SQL, the backbone of modern finance systems. Whether you are auditing invoices, tracking payment balances, or hunting for data anomalies, SQL is your new best friend. Let's get to work!

A. Learning objectives

This lab covers key outcomes from the SDS syllabus areas C2 & D2. By the end of this session, you'll be able to:

- **Apply SQL** across diverse business contexts.
- **Prepare and clean data** using both Python and SQL.
- **Design impactful features** to enhance your datasets.
- **Master data transformations** and conditional logic.

B. Why these skills matter

SQL allows finance professionals to extract, audit, and summarise data with precision. You will use these skills to:

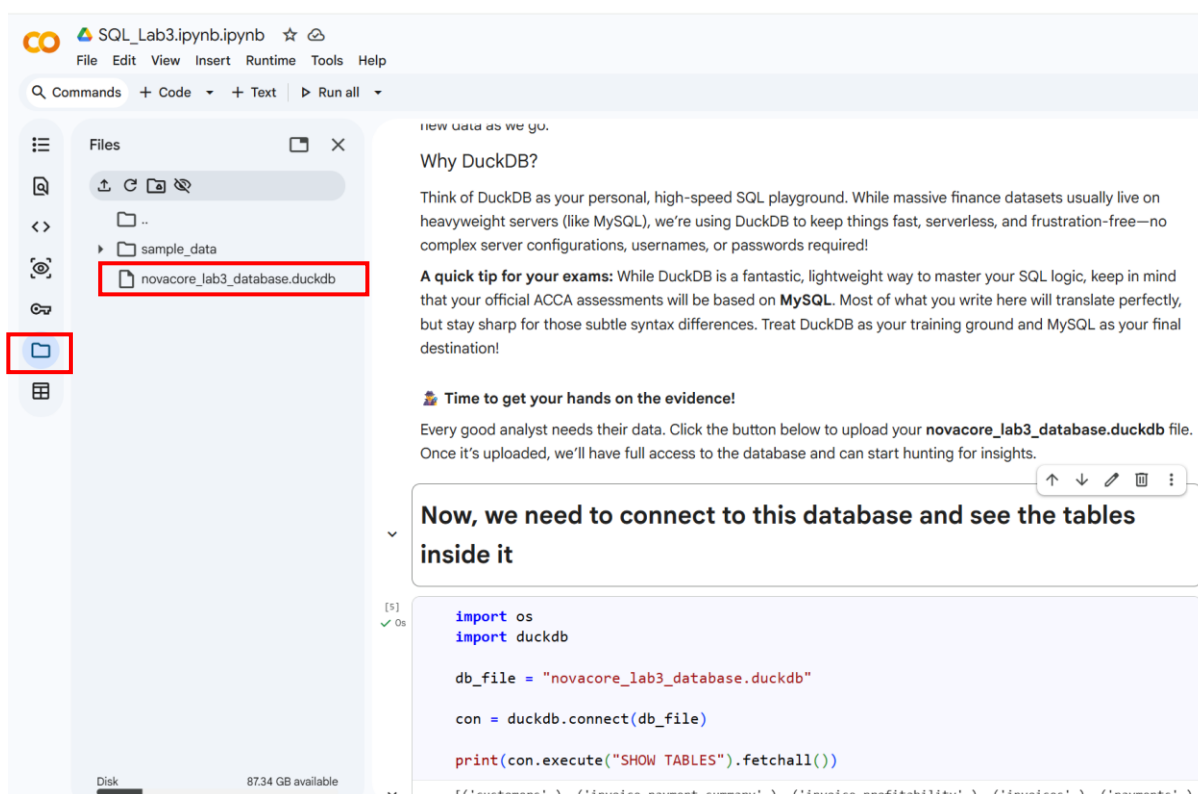
- Review sales invoices and analyse outstanding balances.
- Identify overdue payments and detect duplicate records.
- Build management reports and support critical audit testing.

C. Expected completion time

1–2 hours

2. Setup: Getting Your Engines Running

Once the upload is successful, you'll be able to see the database in your files folder.



The screenshot shows the JupyterLab interface. On the left, the 'Files' sidebar is open, displaying a file explorer. A red box highlights the 'novacore_lab3_database.duckdb' file in the file list. Another red box highlights the 'upload' icon in the sidebar. The main notebook area shows the title 'SQL_Lab3.ipynb.ipynb' and a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu bar is a search bar and a 'Run all' button. The notebook content is titled 'Why DuckDB?' and includes a section on how to connect to the database.

Why DuckDB?

Think of DuckDB as your personal, high-speed SQL playground. While massive finance datasets usually live on heavyweight servers (like MySQL), we're using DuckDB to keep things fast, serverless, and frustration-free—no complex server configurations, usernames, or passwords required!

A quick tip for your exams: While DuckDB is a fantastic, lightweight way to master your SQL logic, keep in mind that your official ACCA assessments will be based on **MySQL**. Most of what you write here will translate perfectly, but stay sharp for those subtle syntax differences. Treat DuckDB as your training ground and MySQL as your final destination!

Time to get your hands on the evidence!

Every good analyst needs their data. Click the button below to upload your **novacore_lab3_database.duckdb** file. Once it's uploaded, we'll have full access to the database and can start hunting for insights.

Now, we need to connect to this database and see the tables inside it

```
[5]
✓ Os
import os
import duckdb

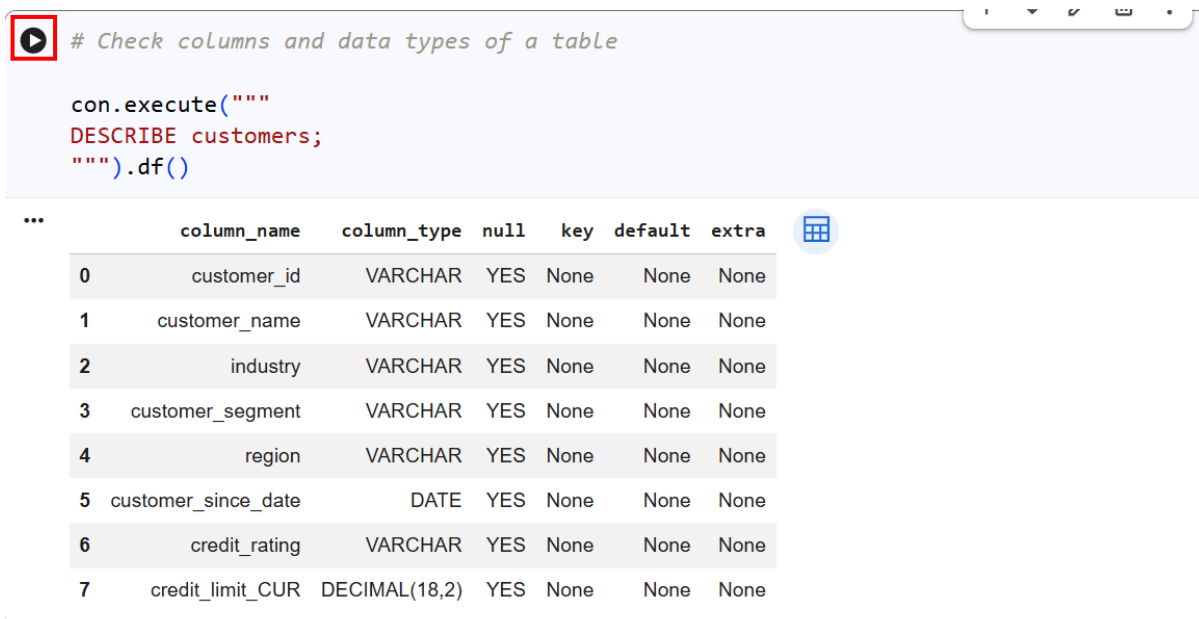
db_file = "novacore_lab3_database.duckdb"

con = duckdb.connect(db_file)

print(con.execute("SHOW TABLES").fetchall())

[('customers'), ('invoice payment summary'), ('invoice profitability'), ('invoices'), ('payments')]
```

3. Follow the instructions in the notebook and run each cell in the order provided by clicking on the run cell icon .



The image shows a Jupyter Notebook interface. At the top, there is a code cell with a red square icon containing a play button. The code in the cell is: `# Check columns and data types of a table`
`con.execute("""`
`DESCRIBE customers;`
`""").df()`

Below the code cell, the output is displayed as a table with 8 rows and 7 columns. The columns are labeled: `column_name`, `column_type`, `null`, `key`, `default`, and `extra`. The rows represent the columns of the `customers` table.

	column_name	column_type	null	key	default	extra
0	customer_id	VARCHAR	YES	None	None	None
1	customer_name	VARCHAR	YES	None	None	None
2	industry	VARCHAR	YES	None	None	None
3	customer_segment	VARCHAR	YES	None	None	None
4	region	VARCHAR	YES	None	None	None
5	customer_since_date	DATE	YES	None	None	None
6	credit_rating	VARCHAR	YES	None	None	None
7	credit_limit_CUR	DECIMAL(18,2)	YES	None	None	None

Optional- Save Your Own Working Copy

- Once the notebook loads, you can create a private version to save your progress.
- In the top menu, select File → Save a copy in Drive.

8. What to do if you cannot access a tool

If you encounter technical issues with your environment (such as network restrictions, browser limitations, or inability to run specific cells), you can still review the material in a read-only format.

You may use the Virtual Walkthrough included in this lab package. The walkthrough allows you to follow the practical process and review expected outputs without completing the activity directly in a third-party tool.

Access: You can locate this resource in your lab package under the filename **Lab_3_Readonly.pdf**

9. Troubleshooting Guide

1. Database file cannot be found

Issue: `FileNotFoundError` when connecting to the database.

Fix: Ensure the `.duckdb` file is located in your notebook's working directory, or update the file path in your `duckdb.connect()` command.

2. Google Colab is unavailable

Issue: You cannot access Google Colab to run Lab 3.

Fix: Open [Try Jupyter](#) and select **JupyterLab**. Upload the notebook and database in

the required folder structure, then open the notebook and run each cell in order using **Shift + Enter**.

3. Unable to connect to the database

Issue: `duckdb.ConnectionException` or "File busy" error.

Fix: Another connection to the database may still be active. Restart your notebook kernel or session, then reconnect to the database.

4. Table does not exist

Issue: Catalog Error: Table with name [...] does not exist.

Fix: Check that you are connected to the correct database file. You can verify the available tables by running:

```
con.execute("SHOW TABLES").fetchall()
```

5. SQL statement fails due to data type errors

Issue: Errors when inserting or updating data.

Fix: Ensure the values match the table's data types. To view a table's structure, run:

```
DESCRIBE table_name;
```

6. Changes are not saved

Issue: Tables or records disappear after restarting the notebook.

Fix: Make sure you are connected to the provided `.duckdb` database file rather than an in-memory database (`:memory:`).

7. Notebook becomes slow or unresponsive

Issue: Large queries take a long time to display.

Fix: Avoid retrieving entire tables. Instead, use `LIMIT` or summarise the data with functions such as `COUNT()`, `SUM()`, or `AVG()`.

Example:

```
SELECT *  
FROM transactions  
LIMIT 50;
```

8. Need to check the database contents

Issue: You are unsure which tables or columns are available.

Fix: Use the following commands:

```
con.execute("SHOW TABLES").fetchall()  
  
DESCRIBE table_name
```